Software Construction and Development

Mid-Term Project Proposal (2026, 5th Semester)

Project: NHRX — National Health Record Exchange

Project Proposal and Introduction of the Project

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PART-I

1. Introduction and Background of the System

The **NHRX – National Health Record Exchange** is a software-based system designed to manage patient health information through a centralized database. It will allow authorized healthcare staff to **store, update, search, and retrieve** patient records from a single platform, including **patient details, medical history, diagnoses, allergies, medications, appointments, and test records**.

The system aims to reduce problems associated with **paper-based and scattered records** by providing organized and easily accessible information. It will connect participating healthcare facilities, allowing authorized users to access relevant patient information when required and helping maintain **continuity of records** when patients receive healthcare services from different hospitals or clinics.

NHRX will support users such as **doctors, nurses, laboratory staff, and system administrators**, with access based on their roles and responsibilities. **Role-based access control** will protect sensitive health information, while **audit mechanisms** will provide traceability of important record activities. The system will provide a **structured, secure, and scalable platform** for managing patient information and supporting secure information exchange between participating healthcare institutions.

Summary: NHRX aims to reduce reliance on paper-based and isolated records while making relevant patient information more **organized, accessible, secure, and easier to exchange**.

PART-II

2. Problem Statements

1. Patient records may be stored in **paper files, spreadsheets, or separate digital systems**.
2. Finding previous medical information can be **time-consuming** when records are stored separately.
3. Manual record management can cause **duplicate, incomplete, outdated, or inconsistent data**.
4. Manual data entry increases the risk of **errors in patient records**.
5. Preparing healthcare reports manually can be **difficult and time-consuming**.
6. Sensitive patient information may lack **proper access control**.
7. A **centralized system** is required to securely store, update, search, and retrieve patient information while controlling access and modifications.
8. There is no **shared system for controlling access to patient data across hospitals**.
9. Storing sensitive medical data in **unprotected form** creates security risks when shared across hospitals.
10. Without proper **decrypt-on-read mechanisms**, decrypted data may remain exposed after viewing.
11. Generating **nationwide healthcare statistics and reports** is difficult when data is stored in separate systems.
12. Patient records continuously grow, requiring **efficient storage and data management**.
13. Different hospitals may use **different data formats and standards**, making information exchange difficult.

Part-III

3. Users of the System

The proposed system will have **five main types of users**:

3.1 Administrator

The administrator will manage **system users, roles, permissions, and basic system settings**. The administrator will also manage **database backups and system maintenance**.

3.2 Doctor / Healthcare Provider

Doctors or healthcare providers will access patient records and manage medical information such as **medical history, diagnoses, medications, allergies, and test results**, according to their permissions.

3.3 Receptionist / Healthcare Staff

Receptionists or authorized healthcare staff will manage **patient registration, basic patient information, and appointments** according to their assigned permissions.

3.4 Pharmacist

Pharmacists will access relevant **medication and prescription information**, manage prescribed medicines, and update medication-related records according to their permissions.

3.5 Health Minister

The Health Minister will access a **dashboard containing nationwide healthcare statistics and reports**, such as patient counts, disease trends, hospital statistics, and other aggregated health information to support **healthcare monitoring.**

3.6 Patient

Patients will be able to view their **personal information, medical history, diagnoses, allergies, medications, test results, and appointments.** Patients will only have access to their own records and will not be allowed to modify medical information entered by authorized healthcare staff.

Part-IV

4. Business Objectives

The major business objectives of the proposed system are:

1. Centralize patient health information in a structured database.
2. Reduce reliance on manual and paper-based records.
3. Provide quick access to authorized patient information.
4. Improve the organization of medical records.
5. Reduce duplicate patient records.
6. Improve the accuracy and consistency of stored information.
7. Provide controlled access to sensitive healthcare information.
8. Simplify appointment management.
9. Support basic healthcare reporting.
10. Reduce the time required to search and manage patient records.
11. Provide a scalable structure for future expansion.

Part-V

5. Basic Functional Requirements

5. System Requirements

Below are some of the functional requirements which the system shall implement:

1. Patient Management – Register, update, search, and view patient records.
2. Role-Based Access – Provide different permissions for Admin, Doctor, Receptionist, Pharmacist, and Health Minister.
3. Medical Records – Manage diagnoses, medical history, allergies, medications, and test results.
4. Appointment Management – Create, update, cancel, and view patient appointments.

5. Prescription Management – Doctors can create prescriptions and pharmacists can view and process them.
6. Audit Logs – Record important activities such as patient record creation, updates, and access.
7. Healthcare Dashboard – Display statistics such as patient counts, diseases, appointments, and hospital data.
8. Secure Authentication – Provide login, password hashing, and secure access to the system.

Part-VI

6.Technologies Used in Project

The NHRX system will use the following technologies for development, database management, and deployment:

Technology	Purpose
Next.js / React	Build the frontend and user interface
Tailwind CSS	Style and create responsive layouts
ShadCN / Material UI	Provide ready-made UI components
Boxicons	Provide icons for the interface
Fetch / Axios	Connect the frontend with the backend API
Node.js	Run JavaScript on the server
Express.js	Build backend REST APIs and application logic
Prisma	ORM for database operations

PostgreSQL	Store patient and system data
AWS	Database hosting and application deployment
Postman	Test and verify API endpoints

Part-VI

7. System Interaction

User → Next.js/React → REST API → Node.js/Express → Prisma → PostgreSQL (Database)

The system will be deployed on **AWS**, which will provide the required cloud infrastructure for hosting the application and database.

Part-VII

8.Complex Computing Problems

1. Role-Based Access Control (RBAC)

Justification:
NHRX contains sensitive patient information, so users should only access data relevant to their responsibilities.

Key Points:
1. Assign permissions based on user roles.
2. Doctors, nurses, and administrators have different access levels.
3. Apply the principle of least privilege.
4. Prevent unauthorized access to patient records.

Objective:
Ensure that only authorized users can access the appropriate patient information.

2. Encryption and Decryption

Justification:
Patient data such as CNIC, diagnoses, and medical history must remain protected even if unauthorized users gain access to stored data.

Key Points:

1. Encrypt sensitive data before storage.
2. Decrypt data only when required by an authorized process.
3. Use TLS to protect data in transit.
4. Secure and regularly manage encryption keys.
5. Encrypt database backups.

Objective:
Protect patient data from unauthorized disclosure.

3. Emergency Access

Justification:
During medical emergencies, healthcare professionals may need immediate access to critical patient information even when normal access procedures cannot be completed.

Key Points:

1. Provide controlled emergency access.
2. Allow access to only necessary medical information.
3. Require an emergency reason or justification.
4. Record every emergency access attempt.
5. Review emergency access through audit logs.

Objective:
Balance **patient privacy** with **timely emergency treatment**.

Overall Security Approach

RBAC → Controls who can access data
Encryption → Protects the data
Emergency Access → Ensures critical data remains available when needed

Part-IX

9.Benchmarking

Platform / System	Medical Record Storage	Role-Based Access Control	Cross-Hospital Data Sharing	Patient Consent Control	Pharmacist & Receptionist Views	Health Minister Analytics Dashboard
Standard Hospital Software (e.g., local clinic app)	Yes	No	No	No	No	No
Traditional Electronic Health Records (EHR)	Yes	Yes	No	No	No	No
Online Pharmacy Apps	No	No	No	No	Yes	No
Our NHRE Project (Python Based)	Yes	Yes	Yes	Yes	Yes	Yes